Temporal 2D Swimmer Pose Estimation with Anatomical Constraints and Synthetic Data

Alexandre Gachenot[1], Vincent Lepetit[2], Rémi Carmigniani[1]

Abstract—Quantitative analysis of swimmer kinematics, such as instantaneous velocity and stroke frequency, is crucial for athletic training, yet automated underwater pose estimation is challenged by occlusions, large field of view, high-velocity movements, and multi-swimmer interference. In this work, we propose an end-to-end deep learning framework designed to extract 2D keypoint coordinates from calibrated multi-camera swimming pools. Utilizing a frozen self-supervised DINO ConvNeXt Large backbone, we deploy a lightweight Temporal U-Net decoder (<7M parameters) that aggregates hierarchical features across multiple frames to maintain temporal consistency. To mitigate anatomical left-right and top-bottom joint confusions, we introduce a Target-Weighted Anatomical Loss incorporating pairwise joint exclusion penalties and a spatial edge penalty. Furthermore, we generate procedurally randomized 3D synthetic sequences in Blender to address data scarcity. Our evaluations across four different swimming pools demonstrate that our proposed framework achieves a global Root Mean Square Error (RMSE) of 9.2 ± 0.2 pixels (at a scale of 50 px/m), with joint-specific errors contextualized at human scale via a canonical Vitruvian representation. The incorporation of synthetic data significantly improves zero-shot generalization to unseen pools, reducing the RMSE on out-of-distribution facilities by up to 4.8 pixels. The synthetic dataset and its generation pipeline are publicly available at https://github.com/AlexandreGac/temporal-swimmer-pose.

Index Terms—Pose Estimation, Swimmer Biomechanics, Sim2Real Adaptation, Temporal Fusion, Deep Learning in Sports.

1 Introduction

Athletic performance optimization in professional swimming relies increasingly on precise, quantitative biomechanical feedback. To help swimmers minimize hydrodynamic drag and maximize propulsion, coaches and sports scientists require precise kinematic indicators, including joint angles, instantaneous velocities, and fluid-structure interaction metrics such as active drag and added mass effects [1]. Traditional methods, such as manual video digitization or passive marker-based motion capture, are either too slow or physically interfere with the swimmer's natural hydrodynamics by altering the boundary layer of water flow. Automated markerless pose estimation offers a non-invasive alternative, enabling continuous tracking of athletes in standard training setups.

However, underwater imaging in athletic swimming pools presents severe computer vision challenges. Swimmers undergo high-velocity displacements where body parts are frequently occluded by air-bubbles, splashing, or self-occlusion. Symmetrical joint confusion (e.g., confusing left and right wrists) and distal joint confusion (e.g., confusing elbows and knees) are highly prevalent. Furthermore, real-world swimming lanes often contain high visual noise, such as background swimmers, lane markers, and floor designs, which frequently trigger false positive joint detections. Finally, deep learning models are notoriously prone to overfitting to the specific visual properties of their training pools, leading to poor generalization in unseen facilities.

To address these limitations, we propose an integrated 2D pose estimation pipeline designed for robust swimmer tracking. We leverage a frozen self-supervised DINOv3 [2] ConvNeXt Large [3] backbone to extract deep spatial features, which is followed by a lightweight Temporal U-Net [4] decoder that aggregates temporal contexts to resolve transient occlusions. To restrict detections to the targeted athlete and resolve limb ambiguity, we introduce a Target-Weighted Anatomical Loss that couples positive target weighting with pairwise joint-exclusion constraints and a spatial edge penalty. Finally, to bypass the scarcity and high cost of annotated real-world underwater data, we generate procedurally randomized 3D synthetic sequences in Blender [5] to facilitate robust simulation-to-reality (Sim2Real) domain adaptation [6].

Our main contributions are:

- A lightweight Temporal U-Net decoder (< 7M parameters) designed to fuse sequential multi-frame feature maps, preserving temporal consistency.
- A unified Anatomical Hybrid Loss consisting of target-weighted keypoint regression, joint exclusion terms, and a spatial edge penalty.
- An exhaustive Sim2Real adaptation study leveraging 3D synthetic swimmers procedurally randomized in Blender, demonstrating a significant improvement in zero-shot generalization across swimming facilities.
- A Vitruvian-based canonical error visualization that maps joint localization errors onto a standardized human-scale template.

2 Related Work

This section reviews the evolution of human pose estimation (HPE) across three pillars relevant to our approach: aquatic-specific pose estimation, temporal coherence in dynamic athletic videos, and the bridging of the simulation-to-reality (Sim2Real) gap through synthetic data.

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2.1 Underwater HPE and Swimming Kinematics

Quantitative swimming biomechanics historically relied on marker-based optical motion capture systems adapted for aquatic environments, utilizing waterproof passive retroreflective markers or active LED tags synchronized with specialized underwater camera arrays [7]. While these optoelectronic systems are widely considered a gold standard for kinematics, they present operational and hydrodynamic drawbacks in swimming applications.

First, physical markers placed on the athlete's skin alter the natural boundary layer of the water flow. In a dedicated hydrodynamic study, Olstad et al. [8] quantified this phenomenon, demonstrating that wearing standard 19-mm spherical motion-capture markers increases a swimmer's passive drag coefficient by 7% to 10%, thereby distorting realistic biomechanical performance. Second, underwater adhesives can fail during high-velocity movements, and light refraction and surface reflections around physical markers can degrade optical tracking accuracy [7]. Lastly, these systems require extensive calibration protocols and are typically restricted to dedicated flumes or research pools, limiting their use in standard training sessions [9].

To address these limitations, marker-less human pose estimation (HPE) has emerged as a non-invasive alternative. Early marker-less methods relied on manual digitization or key-frame detection to extract macroscopic parameters such as stroke frequency [10]. More recently, deep convolutional neural networks (CNNs) have been introduced to continuously regress 2D coordinate joint heatmaps directly from video frames [7], [9]. For instance, Einfalt et al. [9] proposed an activity-conditioned pose estimation framework tailored to athletic swimming analysis.

Despite these advancements, existing marker-less aquatic models face two main practical limitations:

1) **Overfitting to specific pool facilities**: Most published swimming HPE architectures are trained and evaluated on video sequences captured within a single swimming facility [7], [9]. Consequently, these models tend to correlate joint predictions with localized features such as specific tiling patterns, line markings, or lighting configurations, leading to marked performance drops when deployed in unseen facilities. Our framework addresses this challenge by training across heterogeneous environments (pools #1 and #2) combined with a procedural synthetic pool, and evaluating zero-shot generalization across four distinct swimming facilities.
2) **Susceptibility to background clutter and multi-swimmer interference**: Most deep-learning architectures for swimming are benchmarked under isolated conditions with a single swimmer per lane. In regular training sessions, however, athletes frequently share lanes or swim in adjacent lanes, introducing background visual distractors. Without explicit spatial regularization, networks often produce false-positive detections on non-target swimmers.

2.2 Temporal Modeling for Dynamic Video HPE

Single-frame pose estimators, such as HRNet [11] or standard ConvNeXt baselines, process frames independently. When applied to dynamic athletic movements, independent frame-by-frame inference frequently results in temporal jitter and limb identification errors. In general 2D pose estimation, Ronchi and Perona [12] demonstrated that left-right symmetry inversions and false positive swaps account for a major fraction of total localization errors. To penalize such implausible anatomical predictions, early deep learning frameworks coupled convolutional feature extractors with spatial graphical models [13] to enforce structural pairwise priors between connected joints.

To enforce temporal consistency across sequential frames, temporal architectures leverage multi-frame context. Standard approaches rely on 3D Convolutional Networks (3D-CNNs) or recurrent models like LSTMs to model spatial-temporal representations [14]. However, these architectures increase model complexity and can blur high-resolution spatial features essential for precise coordinate regression.

An alternative strategy relies on feed-forward multi-frame feature aggregation. As demonstrated by architectures like PoseWarper [15], aggregating feature representations across a compact temporal window resolves short-term occlusions while maintaining computational efficiency. Our proposed `TemporalUNetReduced` follows this principle, limiting the parameter footprint (<7M parameters) via early channel bottleneck convolutions at multiple feature scales, retaining fine spatial resolution while enforcing temporal smoothness.

2.3 Synthetic Data and Sim2Real Transfer in Sports

Annotating high-resolution athletic pose datasets is labor-intensive and susceptible to manual labeling variability. To mitigate real-world data scarcity, synthetic datasets generated with 3D computer graphics have gained traction. Foundational work on the SURREAL dataset [16] showed that models trained on synthetic human meshes (SMPL) transfer effectively to real human analysis tasks. In sports biomechanics, datasets such as SportsPose [17] have combined motion capture with graphics engines to model complex movements. In swimming specifically, SwimXYZ [18] established a large-scale synthetic benchmark of parametric swimming motions, which has enabled subsequent deep learning studies on swimming style and motion classification [19].

Nevertheless, models trained exclusively on synthetic imagery frequently face a performance drop when applied to real data, known as the Sim2Real gap [20]. Rather than pursuing exact visual photorealism, domain randomization formalized by Tobin et al. [6] demonstrates that exposing networks to wide procedural variations in textures, camera viewpoints, and lighting during training encourages the model to learn invariant representations that generalize zero-shot to real environments.

Our approach applies this paradigm to competitive swimming: we generate procedurally randomized 3D synthetic sequences in Blender with variations in skin tones, swimwear, camera perspectives, and secondary swimmers, and combine them with an anatomical hybrid loss that penalizes symmetric and proximal joint swaps. This setup regularizes the network and improves zero-shot transfer across distinct swimming facilities.

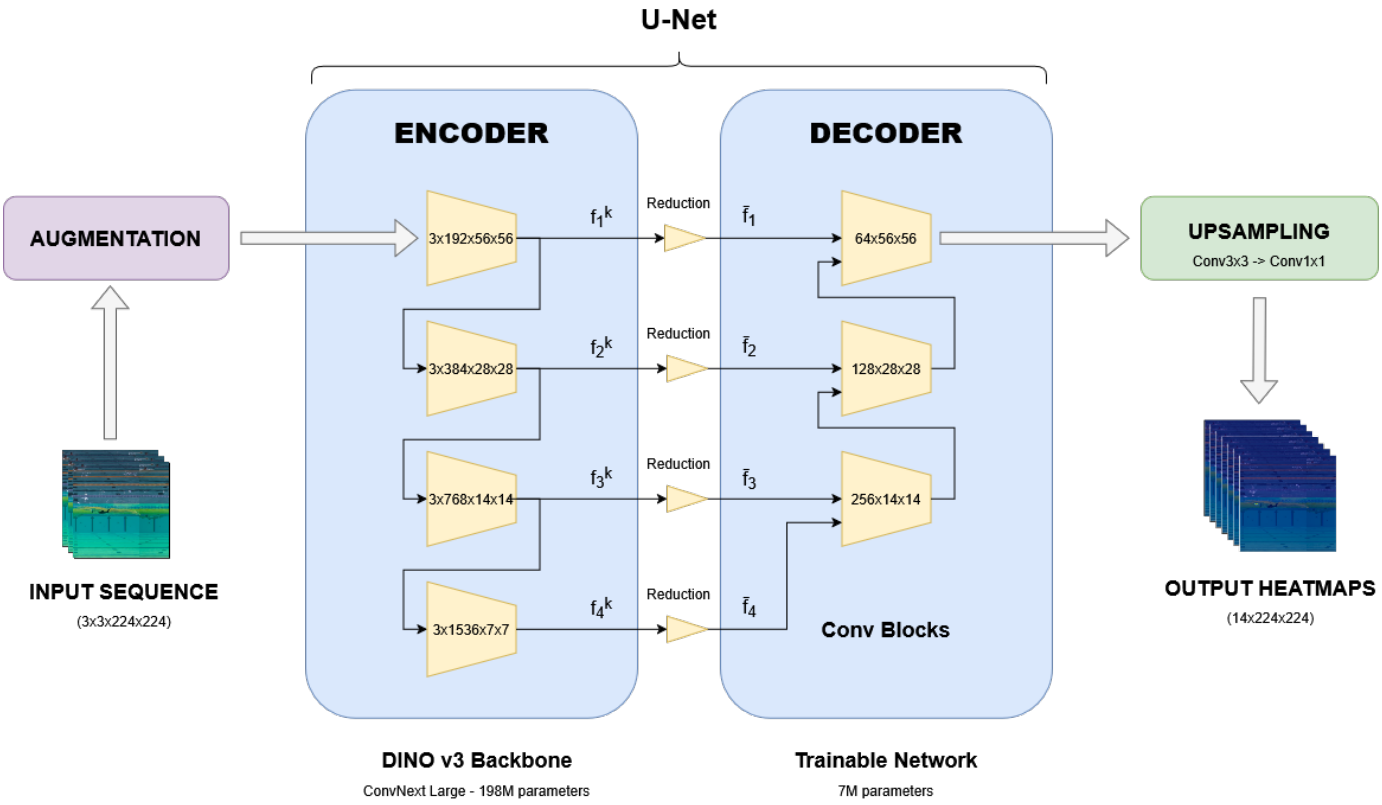


Fig. 1: Overview of the proposed U-Net architecture. For three successive temporal frames, hierarchical features are extracted using the DINOv3 [2] frozen self-supervised backbone. Multi-scale feature maps are fused along the temporal dimension using 1×1 bottleneck convolutions to compress temporal redundancies, before being progressively decoded to produce keypoint heatmaps.

3 Methodology

The proposed framework is designed to estimate 2D joint coordinates of competitive swimmers under complex aquatic conditions. Rather than relying on heavy architectural modifications, our system leverages a design combining frozen deep representations, temporal context fusion, domain-specific randomization, and physically-informed anatomical losses.

3.1 Pipeline Overview and Spatial Tracking

Let $I_t \in \mathbb{R}^{H_0 \times W_0 \times 3}$ denote the video frame captured at time step t by a synchronized and calibrated underwater camera array. Within each camera view, intrinsic parameters and extrinsic projection matrices map 2D pixel coordinates (x, y) directly to real metric distances on a calibrated plane, corresponding to an average spatial resolution of approximately $50\,\text{px/m}$ ($1\,\text{px} \approx 2\,\text{cm}$) at the swimmer's plane of motion.

To focus the model on the athlete, we crop the original frame around the swimmer's running joint centroid $(\bar{x}_t, \bar{y}_t)$. This produces a temporal sequence of super-crops of size 300×300 pixels, sampled with a time step Δt corresponding to a stride of two frames (approximately $0.04\,\text{s}$):

$$\mathbf{X}_{\text{super}} = \{X_{t-\Delta t}, X_t, X_{t+\Delta t}\} \in \mathbb{R}^{3\times 300\times 300\times 3} \tag{1}$$

On the GPU, these super-crops are dynamically cropped to the final network resolution of 224×224 pixels using a randomized translation offset $\delta \sim \mathcal{U}(-15, 15)$ pixels. This introduces spatial translation invariance and prevents the network from overfitting to centered joint distributions.

3.2 Frozen Representation and Temporal Decoder

As illustrated in Fig. 1, we utilize a pre-trained, self-supervised DINOv3 ConvNeXt Large backbone [2] to extract deep spatial representations. To prevent the network from overfitting to the localized color gradients of the training pools, the backbone parameters $\mathbf{\Theta}_{\text{backbone}}$ are frozen during optimization:

$$\nabla_{\mathbf{\Theta}_{\text{backbone}}} \mathcal{L} = 0 \tag{2}$$

For each frame $k \in \{t-\Delta t, t, t+\Delta t\}$, we extract hierarchical, multi-scale feature maps from four intermediate stages of the backbone, denoted as f_1^k, f_2^k, f_3^k, and f_4^k respectively.

To resolve transient occlusions caused by water splashing and bubbles, we employ a lightweight decoder ($\sim$ 7M parameters). Rather than processing temporal sequences through computationally expensive 3D convolutions, the temporal dimensions are compressed early. For each scale $i \in \{1, 2, 3, 4\}$, we first concatenate the feature maps along the channel axis to form:

$$F_i^{\text{cat}} = \left[f_i^{t-\Delta t} \parallel f_i^t \parallel f_i^{t+\Delta t} \right] \tag{3}$$

We then apply a 1×1 bottleneck convolution to compress the temporal redundancies into a unified map $\bar{f}_i$:

$$\bar{f}_i = \text{ReLU}\left(\text{BatchNorm}\left(\text{Conv}_{1\times 1}\left(F_i^{\text{cat}}\right)\right)\right) \tag{4}$$

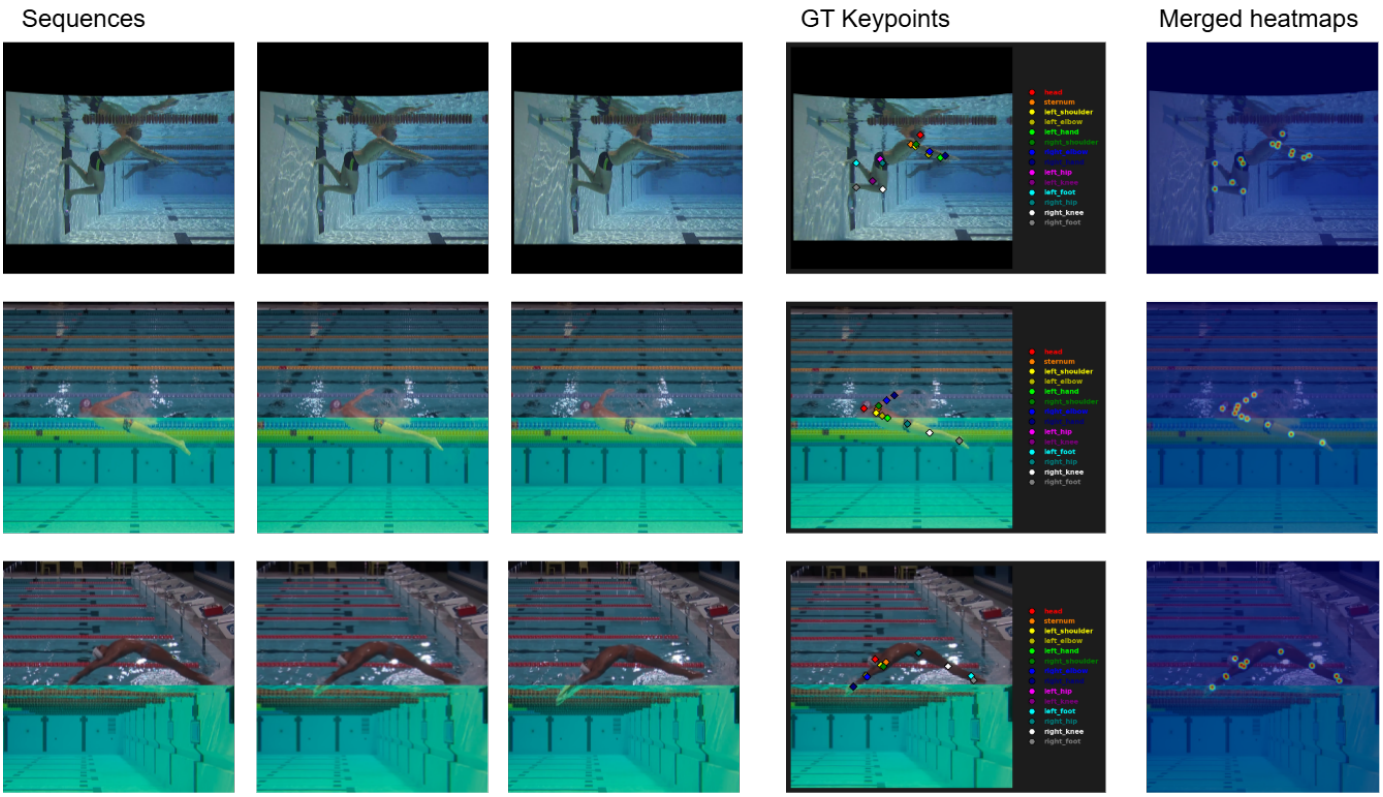


Fig. 2: Representative sample from the training dataset (comprising only pools #1 and #2). The panels illustrate the raw sequence frames, ground-truth keypoint locations, and a target heatmap generated by compiling target joints using a pixel-wise maximum operation.

These compressed feature maps are progressively decoded using bilinear upsampling and Decoder Blocks connected via skip connections. The final decoder layer outputs the continuous heatmap volume $\hat{\mathbf{Y}} \in \mathbb{R}^{224\times224\times C_{\text{kp}}}$, where C_{kp} represents the number of target keypoints ($C_{\text{kp}} = 14$).

3.3 Data Augmentation

To train models capable of generalizing to unseen aquatic facilities, we deploy data augmentation on the GPU. This pipeline acts as a domain randomization layer to bridge the gap between synthetic data, training pools, and unseen facilities.

To simulate physical variations in swimmer displacement and camera perspective, we apply dynamic affine transformations. This includes random rotation $\theta \sim \mathcal{U}(-30°, 30°)$, random scaling $s \sim \mathcal{U}(0.8, 1.2)$, and horizontal flipping with a probability of 0.5 (incorporating label-swapping of bilateral joint pairs). Random spatial cutout blocks of size 30 to 80 pixels are also introduced to simulate localized visual obstructions.

3.4 Anatomical Hybrid Loss Function

The optimization of the proposed network relies on a multi-objective loss function designed to handle domain-specific challenges, such as background distractors and symmetrical joint confusion:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{kp}} + \lambda_{\text{ex}} \left(\mathcal{L}_{\text{ex_lr}} + \mathcal{L}_{\text{ex_tb}} \right) \tag{5}$$

where $\lambda_{\text{ex}} = 40.0$ represents the joint exclusion penalty weight.

3.4.1 *Keypoint Heatmap Regression with Target Weighting*

We supervise the keypoint heatmap regression using a weighted Mean Squared Error (MSE) loss. To balance the highly skewed background-to-foreground pixel distribution, we integrate a positive target weighting mask $W^{\text{tgt}}_{b,c,y,x}$ directly into the formulation. Let $P_{b,c,y,x}$ and $T_{b,c,y,x}$ denote the predicted and ground-truth heatmaps, respectively. The localized keypoint loss is defined as:

$$\mathcal{L}_{\text{kp}} = s_{\text{kp}} \sum_{b,c,y,x} \left(P_{b,c,y,x} - T_{b,c,y,x} \right)^2 \cdot W^{\text{tgt}}_{b,c,y,x} \cdot W^{\text{sp}}_{b,c,y,x} \tag{6}$$

where $s_{\text{kp}} = \frac{1}{B \cdot C_{\text{kp}} \cdot H \cdot W}$ is a normalization scale factor with batch size $B = 64$, heatmap spatial dimensions $H = W = 224$, and target keypoints $C_{\text{kp}} = 14$. The target weight mask $W^{\text{tgt}}_{b,c,y,x}$ is defined as:

$$W^{\text{tgt}}_{b,c,y,x} = \beta \cdot T_{b,c,y,x} + 1.0 \tag{7}$$

with the target weighting coefficient set to $\beta = 80.0$.

3.4.2 *Spatial Edge Penalty for Distractor Suppression*

Underwater scenes are often cluttered by secondary swimmers or lane lines in adjacent lanes, leading to false positive activations near the frame boundaries. To suppress these distractors, we apply a spatial distance-based weight $W^{\text{sp}}_{b,c,y,x}$

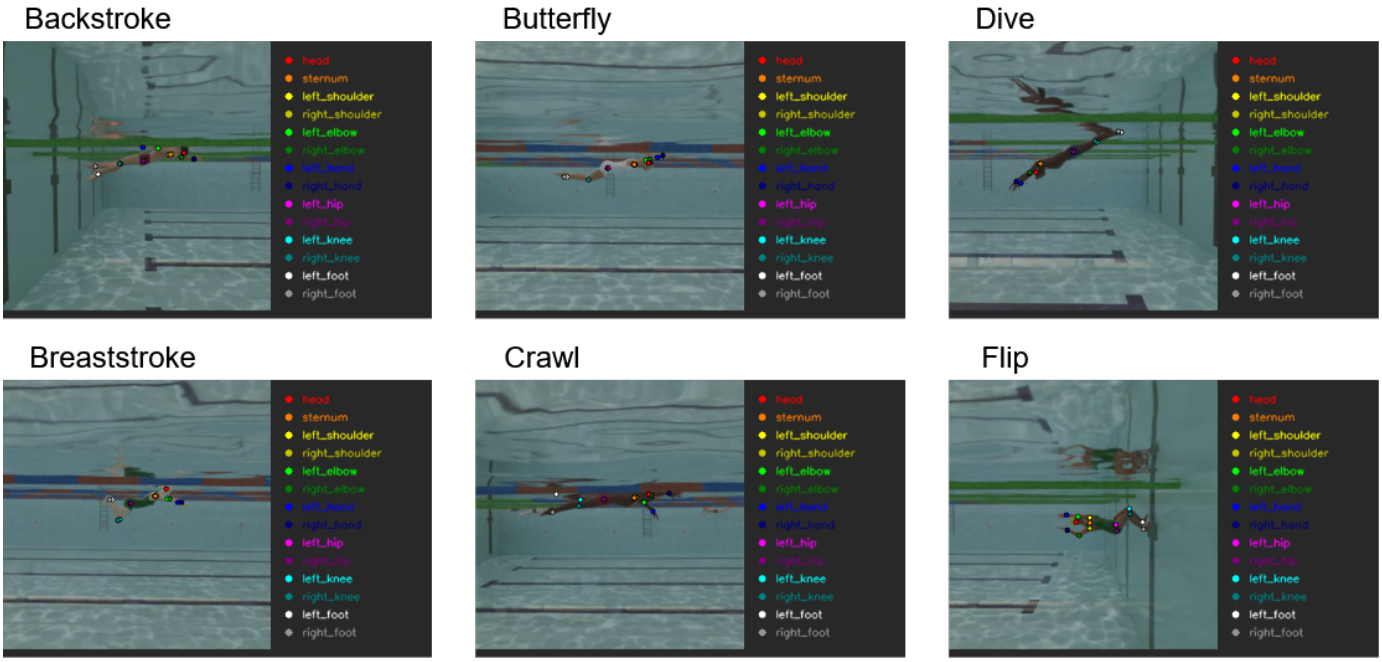


Fig. 3: Procedurally-generated synthetic swimmer sequences from Blender. The synthetic environment models all four competitive strokes, dive starts, and flip turns under domain randomizations, including randomized swimwear and swim cap colors, Fitzpatrick skin tones [21], camera perspective noise ($\pm 1.0°$), and the randomized presence of background distractor swimmers.

that penalizes background activations away from the crop center:

$$W^{\text{sp}}_{b,c,y,x} = 1.0 + M_{b,c,y,x} \cdot D(y,x) \cdot \gamma \tag{8}$$

where $M_{b,c,y,x} = \mathbb{I}(T_{b,c,y,x} < 0.05)$ is the background indicator mask, $\gamma = 10.0$ is the edge penalty coefficient, and $D(y,x)$ represents the normalized Euclidean distance from the crop center:

$$D(y,x) = \frac{\sqrt{(x - W/2)^2 + (y - H/2)^2}}{\sqrt{W/2 \cdot H/2}} \tag{9}$$

3.4.3 Anatomical Joint Exclusion Penalties

Symmetric joint confusion (e.g., left wrist vs. right wrist) and distal joint confusion (e.g., elbows vs. knees) represent common failure modes in heatmap-based pose estimation. To enforce anatomical plausibility and prevent these overlapping activations, we introduce a penalty on concurrent predictions for distinct joint pairs.

Let $\mathcal{P}_{\text{lr}}$ and $\mathcal{P}_{\text{tb}}$ denote the predefined sets of bilateral (left-right) and vertical (top-bottom) joint pairs. For any pair (l, r), we construct the exclusive ground-truth masks representing regions where only one of the joints is present:

$$T^{\text{ex_right}}_{b,l,r,y,x} = T_{b,r,y,x} \cdot (1.0 - T_{b,l,y,x}) \tag{10}$$

$$T^{\text{ex_left}}_{b,l,r,y,x} = T_{b,l,y,x} \cdot (1.0 - T_{b,r,y,x}) \tag{11}$$

The bilateral joint exclusion loss is then defined as:

$$\mathcal{L}_{\text{ex_lr}} = \sum_{(l,r)\in\mathcal{P}_{\text{lr}}} \frac{1}{B \cdot H \cdot W} \sum_{b,y,x} \Big[\Big(P_{b,l,y,x} \cdot T^{\text{ex_right}}_{b,l,r,y,x} \Big)^2 + \Big(P_{b,r,y,x} \cdot T^{\text{ex_left}}_{b,l,r,y,x} \Big)^2 \Big] \tag{12}$$

Similarly, for top-bottom joint pairs $(u, v) \in \mathcal{P}_{\text{tb}}$, the exclusion loss is formulated as:

$$\mathcal{L}_{\text{ex_tb}} = \sum_{(u,v)\in\mathcal{P}_{\text{tb}}} \frac{1}{B \cdot H \cdot W} \sum_{b,y,x} \Big[\Big(P_{b,u,y,x} \cdot T^{\text{ex_bot}}_{b,u,v,y,x} \Big)^2 + \Big(P_{b,v,y,x} \cdot T^{\text{ex_top}}_{b,u,v,y,x} \Big)^2 \Big] \tag{13}$$

where $T^{\text{ex_bot}}_{b,u,v,y,x}$ and $T^{\text{ex_top}}_{b,u,v,y,x}$ are defined analogously.

4 Experimental Setup

This section details the experimental protocol, the composition of both the real-world and procedurally-generated datasets, and the parameters used to train and evaluate the proposed models.

4.1 Real-World Datasets and Annotation Protocol

Our real-world evaluation framework utilizes video sequences captured across four distinct physical swimming facilities. Each facility presents a unique visual environment where water color, clarity, and pool bottom designs (such as tiling patterns and lane markings) vary significantly. Synchronized and calibrated camera projection matrices are used within each camera view to map joint coordinate estimations directly back to a 2D plane within the physical pool coordinates, yielding an effective scale of $\sim$50 px/m across the captured swimming lanes. Notably, the pool #1 dataset results from a dual-camera setup that fuses synchronized aerial and underwater views into a unified perspective.

As shown in Fig. 2, the real-world dataset covers a wide spectrum of visual conditions across these facilities.

TABLE 1: Dataset partitioning across swimming facilities and strokes.

Split / Facility	Freestyle	Backstroke	Breaststroke	Butterfly	Starts / Kicks	Turns	Total
Training Set							
Pool #1	513	615	92	170	1381	154	2925
Pool #2	333	–	–	6	174	–	513
Synthetic	242	242	242	242	24	84	1,076
Validation Set							
Pool #1	53	17	19	33	145	116	383
Pool #2	19	20	–	–	10	–	49
Testing Set							
Pool #1	–	50	81	35	50	–	216
Pool #2	80	24	–	–	23	10	137
Pool #3†	46	–	–	–	9	–	55
Pool #4†	44	–	–	20	31	18	113

† Unseen facilities withheld during training to benchmark zero-shot transfer.

The sequences incorporate a large variety of competitive swimming styles, covering all four competitive strokes (crawl/freestyle, backstroke, breaststroke, and butterfly), and dynamic athletic scenarios, including dive starts, flip turns, and underwater dolphin kicks.

The annotation protocol defines $C_{\text{kp}} = 14$ standard anatomical joints: Head, Sternum, Left/Right Shoulder, Left/Right Elbow, Left/Right Wrist, Left/Right Hip, Left/Right Knee, and Left/Right Ankle. Each sequence was manually annotated (see Fig. 5).

The dataset partitions are structured to evaluate the zero-shot generalization capability of the networks across unseen pool environments, as detailed in Table 1:

- **Training Set**: Comprises real video sequences from pool #1 and pool #2, augmented with procedurally generated synthetic sequences from the virtual pool during Sim2Real transfer experiments.
- **Validation Set**: Contains sequences from pool #1 and pool #2, dedicated to model monitoring, hyperparameter tuning, and checkpoint selection.
- **Testing Set**: Comprises independent test sequences from all four physical facilities (pools #1 to #4). To benchmark out-of-distribution generalizability, pools #3 and #4 are completely withheld from the training phase.

4.2 Procedural Synthetic Data Generation

To address annotated data scarcity and mitigate the high labor cost of manual underwater labeling, we developed a 3D procedural swimming simulation environment utilizing Blender [5] and the Cycles rendering engine, accelerated via NVIDIA OptiX GPU compute frameworks.

The synthetic generation pipeline is structured to produce biomechanically and demographically diverse training samples. The human meshes and skeletal rigs are generated using the MakeHuman Plugin for Blender (MPFB) addon [22], providing anatomical human models for both male and female swimmers. The movement animations are sourced and adapted from biomechanical recordings from SWUM [23], modeling the dynamic motions of crawl, backstroke, breaststroke, and butterfly, as well as athletic starts and flip turns.

The procedural rendering pipeline operates on 16 distinct rigged swimmer meshes along with three active virtual cameras configured dynamically based on the current swim stroke or action. We also apply a multi-layered domain randomization protocol during generation:

- **Demographic and Skin Tone Diversity**: The skin material color of the foreground swimmer is randomized across the 6-level Fitzpatrick skin type scale [21], mapping hex color values from Type I to Type VI.
- **Swimwear and Accessories**: Swimmer models are assigned randomized swimsuit and cap colors selected from 8 contrastive palettes.
- **Multi-Swimmer Interference**: To simulate crowded pool conditions and evaluate distractor suppression, secondary swimmers are spawned in adjacent lanes with a probability of 0.5. These background models undergo independent swimwear and skin tone randomizations to maximize visual diversity.
- **Camera Misalignment and Vibration**: To emulate camera calibration drift, structural vibrations, or underwater current perturbations in real pools, a randomized Euler rotation perturbation is injected into the active camera's orientation matrix (with uniform noise within $\pm 1.0^\circ$ applied to the X, Y, and Z axes) before rendering.
- **Ground-Truth Extraction**: Joint labels are automatically extracted by projecting the 3D coordinate positions of the rig's bone heads (plus custom head and sternum joints) onto the randomized 2D camera plane.

Using this virtual pool, we generated a synthetic dataset containing $1,076$ frames (see Fig. 3). This dataset is mixed into the training partition during Sim2Real transfer experiments to evaluate domain generalization gains.

5 Results and Discussion

This section presents an evaluation of each component in our proposed framework through a multi-run ablation study, followed by an assessment of the model's inference performance, joint-specific error characteristics, and training dynamics. Additional quantitative performance metrics and training dynamics are presented in the Appendix.

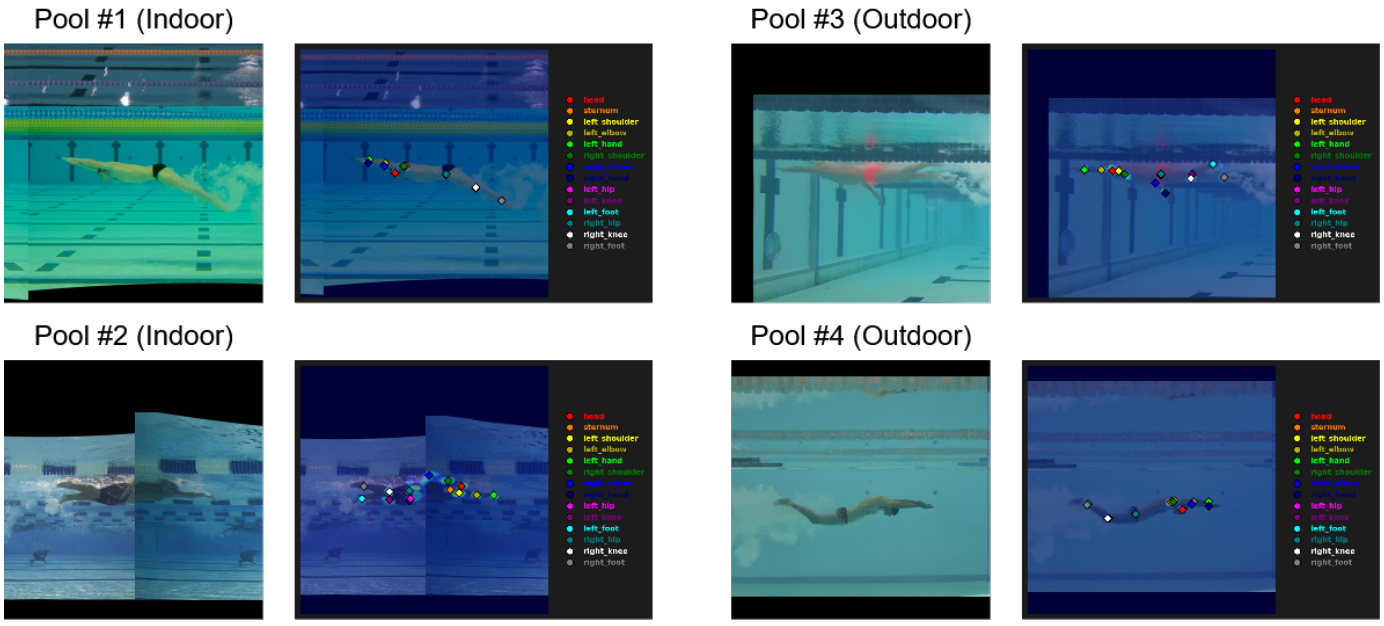


Fig. 4: Inference examples across the four evaluated facilities. Left: raw central frame (time t). Right: predicted keypoint heatmaps generated by pixel-wise maximum fusion of all 14 target channels, overlaid with the 14 keypoint coordinates extracted by localizing the peak activation of each individual heatmap.

5.1 Ablation Study and Performance Analysis

To evaluate the contribution of each algorithmic block, we conducted a systematic ablation campaign over 10 independent runs per configuration. The consolidated mean scores, 95% confidence intervals, and best-performing single runs (Top-1) are reported in Table 2.

5.1.1 Algorithmic Steps of the Ablation Study

The configurations defined in Table 2 correspond to the progressive addition of our proposed methodologies:

- **Baseline (#0)**: Represents a static single-frame model processing the frame at time t. It includes positive keypoint target weighting but excludes temporal aggregation, joint exclusions, spatial edge penalties, and synthetic data. It yields a Global RMSE of 12.47 ± 0.17 pixels, with high generalization errors on the unseen pools (17.32 on pool #3, 17.10 on pool #4).
- **+ Temporal (#1)**: Incorporates sequential multi-frame context ($t-\Delta t, t, t+\Delta t$) through the lightweight Temporal U-Net decoder. The temporal fusion reduces the Global RMSE to 11.72 ± 0.16 pixels, and achieves the highest localized precision on the majority training pool #1 with 2.43 ± 0.06 pixels.
- **+ L/R Exclusion (#2)**: Introduces the bilateral left-right anatomical joint exclusion loss to penalize overlapping predictions on symmetric limbs. This constraint significantly reduces tracking errors on the minority training pool (pool #2 drops from 14.04 ± 0.36 to 10.96 ± 0.88 pixels).
- **+ T/B Exclusion (#3)**: Extends the joint exclusion constraint to top-bottom anatomical pairs (elbows vs. knees, wrists vs. ankles), helping the model distinguish between upper and lower extremities. This structural regularizer further lowers the Global RMSE to 10.63 ± 0.26 pixels.
- **+ Edge Penalty (#4)**: Activates the spatial edge penalty to suppress false positive background activations near frame boundaries. This spatial boundary guides the network to ignore distractors, yielding a Global RMSE of 10.04 ± 0.24 pixels.
- **+ Synthetic (#5)**: Enriching the training partition with procedurally generated 3D synthetic sequences from Blender delivers our best overall configuration, reducing the Global RMSE to 9.21 ± 0.19 pixels (Top-1: 8.75 pixels) and drastically improving zero-shot generalizability on unseen facilities.

5.1.2 Generalization and Pool #1 Performance Trade-off

We observe that the best localized performance on the majority pool (pool #1) is achieved at step #1 (2.43 ± 0.06 pixels). Introducing anatomical constraints and synthetic domain data in subsequent steps slightly increases localized error to 3.53 ± 0.56 pixels. However, this trade-off is highly beneficial: it significantly increases the model's generalizability on minority pools (pool #2 drops from 14.04 to 9.24 pixels) and unseen zero-shot pools (pool #3 drops from 16.15 to 12.49 pixels, pool #4 drops from 15.97 to 13.62 pixels). This demonstrates that the full framework successfully trades a minor loss in localized precision on the dominant training pool to ensure highly robust out-of-distribution generalizability across heterogeneous pool layouts.

5.2 Qualitative Analysis and Visual Inference

Qualitative inference samples obtained with the proposed model (Config. #5) across the four evaluated facilities are illustrated in Fig. 4. The model consistently localizes the 14

TABLE 2: Ablation Study Results (RMSE in Pixels, Mean ± 95% CI and Top-1 over 10 Runs; nominal scale is ∼50 px/m).

ID	Configuration	Global (Mean)	Global (Top-1)	Pool #1	Pool #2	Pool #3†	Pool #4†
#0	Baseline	12.47 ± 0.17	11.79	3.13 ± 0.31	14.58 ± 0.51	17.32 ± 0.64	17.10 ± 0.45
#1	+ Temp.	11.72 ± 0.16	11.32	$\mathbf{2.43 \pm 0.06}$	14.04 ± 0.36	16.15 ± 0.36	15.97 ± 0.34
#2	+ L/R Excl	11.11 ± 0.46	9.51	6.27 ± 0.69	10.96 ± 0.88	13.04 ± 0.58	16.14 ± 0.92
#3	+ T/B Excl	10.63 ± 0.26	9.90	4.94 ± 0.44	11.54 ± 0.62	12.87 ± 0.38	15.16 ± 0.67
#4	+ Edge Pen	10.04 ± 0.24	9.42	3.46 ± 0.17	10.20 ± 0.40	12.83 ± 0.33	15.30 ± 0.76
#5	+ Synth	$\mathbf{9.21 \pm 0.19}$	**8.75**	3.53 ± 0.56	$\mathbf{9.24 \pm 0.49}$	$\mathbf{12.49 \pm 0.29}$	$\mathbf{13.62 \pm 0.43}$

† Unseen pools withheld to evaluate zero-shot transfer.

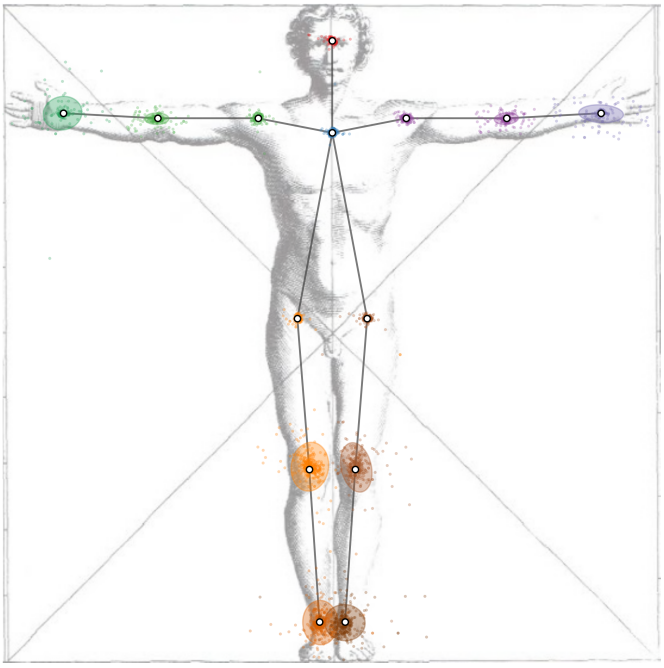


Fig. 5: Spatial error distribution and 95% confidence ellipses (2σ) projected onto the canonical *Homo ad Quadratum* reference pose for the test set, using the global top-1 model (#5). Ground-truth joint locations are indicated by white markers. Predicted coordinates are locally transformed into the coordinate system of each anatomical segment and scaled proportionally to relative segment lengths ($s_k = L_k/R_k$). Errors on proximal joints (head, sternum, shoulders, hips) are compact and isotropic, whereas tracking dispersion increases on distal joints.

anatomical keypoints of the target swimmer across different viewing angles, water turbidities, and surface splashing conditions. In multi-swimmer scenarios, the combination of the spatial edge penalty and target weighting maintains tracking focus on the foreground athlete without generating spurious activations on adjacent lane occupants.

5.3 Spatial Error Distribution on Canonical Representation

To evaluate the spatial distribution and directional tendencies of localization errors independently of swimmer position and camera distance, predictions across the test set are mapped onto a standardized anatomical template (*Homo ad Quadratum*), as shown in Fig. 5. Rather than applying a naive global translation, prediction errors are reprojected within the local kinematic frame of each anatomical segment: trunk keypoints are referenced to the torso centroid, while limb extremities are expressed relative to their parent joints. Each displacement vector is rotated to match the orientation of the canonical segment and scaled by the ratio of canonical-to-observed segment lengths ($s_k = L_k/R_k$), thereby normalizing deviations proportionally to the swimmer's visible body scale.

As shown in Fig. 5, core anatomical joints such as the head, sternum, shoulders, and hips, display minimal coordinate variance, yielding compact and isotropic distributions centered on the ground-truth targets. Tracking dispersion naturally increases toward the extremities. While the overall error distributions remain predominantly circular, a slight longitudinal elongation is noticeable along the segment axes of intermediate joints, particularly for the elbows and knees. This directional bias is less pronounced for the wrists and ankles, where higher velocities, splashing, and localized turbulence produce a larger, more diffuse spatial spread.

5.4 Evaluated Pipeline Alternatives and Negative Results

During framework development, two additional strategies were experimentally assessed but omitted from the final configuration due to the absence of empirical performance improvements:

- **Water-Specific Optical Transformations**: Simulating underwater light attenuation via randomized red-channel attenuation ($\alpha_{\text{red}} \sim \mathcal{U}(0.3, 1.0)$) and box-blur filtering (5×5, $p = 0.3$) did not yield accuracy gains on real test sequences and tended to over-regularize spatial representations.
- **Auxiliary Limb Heatmap Supervision**: Supervising 8 intermediate limb segments via combined segment MSE and Total Variation (TV) regularization failed to improve keypoint localization and slightly increased mean tracking error.

6 Conclusion and Future Work

In this paper, we presented a robust 2D pose estimation framework for the biomechanical analysis of competitive swimmers in calibrated multi-camera pools. Our lightweight

Temporal U-Net decoder, combined with a Target-Weighted Anatomical Loss and spatial constraints, effectively handles severe water splashing, multi-swimmer interference, and transient joint occlusions. Furthermore, our extensive cross-pool evaluations demonstrate that blending procedurally randomized 3D synthetic renders with real-world training frames significantly boosts the zero-shot generalization capabilities of frozen spatial representation backbones on completely unseen facilities.

Several promising avenues exist to extend this work. First, scaling up both the quantitative volume and the physical rendering quality of the procedurally generated synthetic data would allow for a deeper evaluation of whether Sim2Real domain adaptation gains scale continuously with dataset size. Second, while our current model utilizes a localized three-frame temporal window to interpolate transient occlusions, future investigations could explore processing complete, end-to-end video sequences. Integrating long-term sequential modeling architectures—such as recurrent neural networks, state-space models, or temporal transformers—could further enhance temporal tracking consistency.

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References

[1] R. Carmigniani, V. Coloretti, T. Brunel, P. Bosetto, S. Fantozzi, M. Cortesi, and P. Zamparo, "A novel method for assessing added mass in front crawl swimming," *Journal of Biomechanics*, vol. 189, p. 112816, 2025.

[2] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa *et al.*, "DINOv3," *arXiv preprint arXiv:2508.10104*, 2025.

[3] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11 976–11 986.

[4] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. Springer, 2015, pp. 234–241.

[5] Blender Online Community, *Blender - a 3D modelling and rendering package*, Blender Foundation, Stadhoudeskade 112, 1073 PX Amsterdam, The Netherlands, 2018. [Online]. Available: http://www.blender.org

[6] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaytsev, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30.

[7] M. Rossi *et al.*, "Swimmernet: Underwater 2d swimmer pose estimation exploiting fully convolutional neural networks," *Sensors*, vol. 23, no. 4, p. 2200, 2023.

[8] M. Olstad *et al.*, "Automatic 3d motion capture of swimming: Marker resistance," *Journal of Sports Sciences*, vol. 38, no. 10, pp. 1120–1128, 2020.

[9] M. Einfalt, D. Zecha, and R. Lienhart, "Activity-conditioned continuous human pose estimation for performance analysis of athletes using the example of swimming," in *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 446–455.

[10] D. Zecha, T. Greif, and R. Lienhart, "Pose estimation for deriving kinematic parameters of competitive swimmers," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2014.

[11] J. Wang *et al.*, "Deep high-resolution representation learning for human pose estimation," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 43, no. 2, pp. 560–586, 2020.

[12] M. R. Ronchi and P. Perona, "Benchmarking and error diagnosis in multi-instance pose estimation," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 369–378.

[13] J. J. Tompson, A. Jain, Y. LeCun, and C. Bregler, "Joint training of a convolutional network and a graphical model for human pose estimation," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 27, 2014.

[14] X. Deng *et al.*, "Human pose estimation and lstm-based diver heading prediction for auv navigation guidance," *IEEE Access*, vol. 9, pp. 95 214–95 225, 2021.

[15] G. Bertasius, H. Wang, and L. Torresani, "Learning temporal pose estimation from sparsely-labeled videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 8427–8436.

[16] G. Varol, J. Romero, X. Martin, N. Mahmood, M. J. Black, I. Laptev, and C. Schmid, "Learning from synthetic humans," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 109–117.

[17] C. K. Ingwersen *et al.*, "Sportspose: A dynamic 3d sports pose dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2023.

[18] G. Fiche, V. Sevestre, C. Gonzalez-Barral, S. Leglaive, and R. Séguier, "SwimXYZ: A large-scale dataset of synthetic swimming motions and videos," in *Proceedings of the ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG)*, 2023, pp. 1–7.

[19] S. Al-Majnoni *et al.*, "MOAR: A swimmer motion swimming style identification model using deep learning," *Engineering, Technology & Applied Science Research*, vol. 15, no. 1, pp. 19 253–19 259, 2025.

[20] C. Doersch and A. Zisserman, "Sim2real transfer learning for 3d human pose estimation: motion to the rescue," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[21] T. B. Fitzpatrick, "The validity and practicality of sun-reactive skin types I through VI," *Archives of Dermatology*, vol. 124, no. 6, pp. 869–871, 1988.

[22] MakeHuman Community, "MPFB: Human character generator for Blender," https://static.makehumancommunity.org/mpfb.html, 2024.

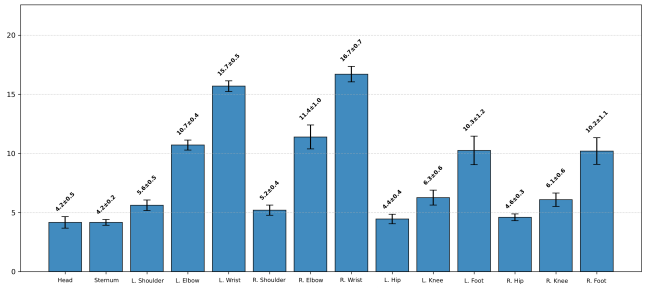
[23] M. Nakashima, "Swimming human simulation model SWUM," http://www.swum.org/, 2023.

APPENDIX

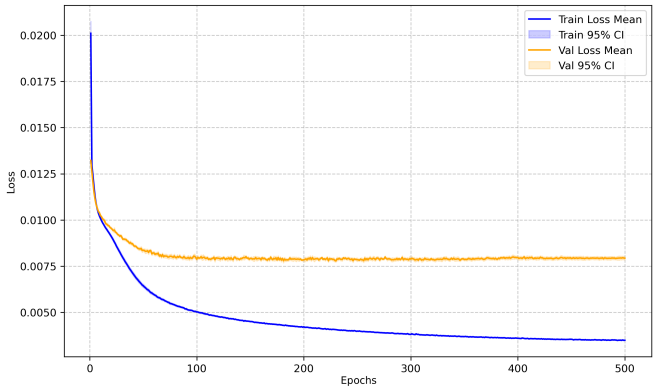
Supplementary Quantitative Metrics, Training Dynamics, and Dataset Release

6.1 Keypoint-Specific Localization Errors

Fig. 6a reports the keypoint-specific Root Mean Square Error (RMSE) computed across the complete test set. Localization error correlates positively with distance from the body centroid: proximal joints (shoulders and hips) maintain an RMSE below 6.0 pixels, whereas distal extremities (wrists and ankles) reach much larger errors. This progression matches the spatial dispersion observed on the canonical representation in Section 5.3, reflecting higher instantaneous velocities and frequent occlusion events at the extremities.



(a) Global keypoint-specific RMSE (in pixels) across the test set. Whiskers indicate 95% confidence intervals across 10 independent evaluation runs.



(b) Training and validation loss convergence across 500 epochs, averaged over 10 runs with shaded 95% confidence intervals.

Fig. 6: Supplementary quantitative evaluations of the proposed framework.

6.2 Training Loss and Optimization Convergence

The convergence profiles for the training and validation losses over 500 epochs are shown in Fig. 6b. The multi-task anatomical objective converges monotonically under the Cosine Annealing learning rate schedule. Validation loss decreases steadily during the initial 100 epochs and asymptotes thereafter, with no evidence of empirical overfitting.

6.3 Open-Source Code and Dataset Release

To facilitate reproducibility and support further research in aquatic pose estimation, the procedural 3D generation pipeline, as well as the associated dataset, is publicly available at https://github.com/AlexandreGac/temporal-swimmer-pose.